

The COVID-19 Exercise

Digital traces, and detecting when an outbreak is taking off

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A different question

ARGO this afternoon asked “**how high is it?**” Early COVID asked a sharper one:

Is it taking off, right now?

Not the level, but the **onset**: is the curve in **exponential growth**? That is the earliest, most actionable signal an outbreak gives.

- Data: **COVID-19 in Washington State**, case counts plus five digital traces.
- Same agent workflow; today the model is a growth detector, not a nowcaster.
- Continues Mauricio’s exercise 3 (`covid_traces_WA.csv`).

Accessing the digital traces

Detecting exponential growth

From growth to an outbreak flag

Multiple traces: early warning

Wrap-up

Accessing the digital traces

Five digital traces, one outbreak

covid_traces_WA.csv pairs the ground truth with the novel streams of the COVID era:

- **Ground truth:** new_cases (positive tests).
- **Traces:** upToDate (clinician search), cdc_ili, Twitter_RelatedTweets, google_fever, Kinsa (smart-thermometer fever), Cuebiq_Mobility.

*The whole course in one file: many independent digital traces, one disease to track.
(Pre-smoothed, so case counts are fractional.)*

Prompt

```
Load covid_traces_WA.csv (daily, Washington State). In a
6-panel plot, show new_cases and the five digital traces
over time. Which traces look like they rise before cases
(leading) and which after (lagging)?
```

- Before any algorithm, form the intuition by eye: where do outbreaks *start*, and do any traces move first?
- Leading traces are the prize: they are what an early-warning system watches.

Detecting exponential growth

What “exponential growth” means, operationally

Exponential growth is a **constant multiplicative factor** each step: today is roughly α times a recent baseline, with $\alpha > 1$.

- Over a sliding 11-day window, regress the **next 10 days on the previous 10** (through the origin, no intercept). The slope is α .
- $\alpha > 1$ means the trace is **growing**; $\alpha < 1$ means it is receding.

Same idea as a log-linear fit: $\alpha \approx e^r$, so doubling time is $\ln 2 / \ln \alpha$. The slope is just an easy, robust way to read it.

Compute the growth factor

Prompt

```
For new_cases, slide an 11-day window: in each window  
regress the last 10 days on the previous 10 with no  
intercept, and store the slope as alpha for that day.  
Then make a 3-panel plot: cases, alpha over time with a  
line at 1, and a 0/1 flag for alpha > 1.
```

Now “is it growing?” is a number you can watch every day, not a judgment call.

From growth to an outbreak flag

Turning the signal into a decision

A single day of $\alpha > 1$ is noise. A **sustained** run is an outbreak.

Prompt

Detect outbreak starts: when $\alpha > 1$ for 10 consecutive days, mark that 10th day as an outbreak start (only the first such day). When $\alpha < 1$ for 10 consecutive days, the outbreak is over and a later run can start a new one. Mark the detected starts on the case-count plot.

A tiny state machine on top of α : growing, active, over, repeat.

It finds the real Washington waves

Run on `new_cases`, the detector flags starts that line up with the actual COVID waves:

2020-03-10 • 2020-06-06 • 2020-09-27 • 2021-03-28 (Alpha) • 2021-07-21 (Delta) •
2021-12-29 (Omicron)

Verify, as always

Do the marks match where *you* said outbreaks start by eye? A detector that fires late, or on every wobble, is worse than useless. Tune the window and threshold and see.

Multiple traces: early warning

Do the traces flag growth before cases?

The point of the digital traces: run the *same* detector on each of them.

Prompt

Repeat the alpha computation and outbreak detection for each digital trace. Redo the 6-panel plot with each trace's outbreak starts marked. For each trace, report how many days before (or after) the case-count outbreak its own outbreak fires.

If enough traces flag growth ahead of cases, you can call an imminent outbreak. That is exactly Mauricio and Kogan's multi-trace early-warning idea.

Wrap-up

What you built

- An **onset detector**: a growth factor α and a small state machine that calls the start of an outbreak, validated against the real WA waves.
- A **multi-trace** version that asks which digital streams lead, the seed of an early-warning system.
- It complements ARGO: ARGO says *how high*, this says *it is taking off*.

Two questions, two tools, both built from simple regressions on novel data.

Looking ahead to Day 3: bring your own problem

Tomorrow you build a situational-awareness product of your own. Start choosing now:

- **A disease:** flu, dengue, RSV, COVID, or something you work on.
- **A place:** your county, your country, Georgia for the capstone framing.
- **A question:** how high (ARGO), taking off (today), or where next (mobility)?

Every skill this week is a Lego brick. Tomorrow you pick which bricks your problem needs. (Mauricio's COVID research talk follows this exercise.)

- Exercise continues Mauricio Santillana's COVID exercise 3 (MIGHTE-1ab/SISMID25); data `covid_traces_WA.csv` (Washington State, daily, 2020–2021), pre-smoothed.
- Traces: UpToDate clinician search, CDC ILI, Twitter, Google “fever”, Kinsa smart-thermometer fever, Cuebiq mobility.
- Multi-trace early warning: Kogan, Santillana et al., *An early warning approach to monitor COVID-19 activity with multiple digital traces*, Science Advances 2021.

Questions?

Repo: `github.com/Shihao-Yang/sismid2026`